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Construction of Consumption Data Optimization Model Based on XGBoost Algorithm

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Abstract

In view of the difficulty in capturing the interactive effects and nonlinear relationships of factors affecting consumer purchase decisions and the imbalance of categories of purchase decision results, this article combines DNN (Deep Neural Network) and XGBoost (Extreme Gradient Boosting) algorithms to study the improvement of consumer purchase decision precision in financial and commercial big data, aiming to better understand consumer needs and improve marketing accuracy. First, transaction records and browsing history are obtained from an e-commerce platform and preprocessed. Then, a DNN multi-layer network structure is used to learn complex nonlinear relationships through nonlinear activation functions. In order to increase the prediction accuracy under unbalanced data, an ensemble of several decision trees is constructed and the parameters and loss functions are changed in conjunction with XGBoost. According to the experimental data, this model's accuracy is 15.8%, 10.5%, 13.0%, 14.4%, and 8.2% higher than the control model's, while its precision is 17.5%, 10.2%, 13.0%, 15.9%, and 10.3% higher. The conclusion shows that the application of DNN and XGBoost algorithms can help improve the understanding of consumer behavior and decision, and can provide new perspectives and basis for corporate marketing.

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1. Introduction

Big data in the commercial and financial domains is growing in size and complexity in tandem with the digital economy's explosive growth. Consumer purchasing behavior is the most important part of an enterprise, and has an important impact on the marketing strategy and competitiveness of the enterprise. How to precisely predict

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consumer behavior is a major challenge currently faced by enterprises [1-2]. Consumer purchasing behavior is a complex process in which multiple factors work together and influence each other [3]. Existing analysis models are difficult to capture it effectively, and the proportion of actual buyers to potential buyers is relatively small. The classification imbalance of consumers in purchasing decisions is relatively serious, which further restricts the prediction precision.

With the in-depth development of the field of machine learning (ML), DNN (Deep Neural Network) and XGBoost (Extreme Gradient Boosting) algorithms have made great progress. DNN has shown strong advantages in feature extraction and nonlinear relationship modeling, while XGBoost has strong capabilities in structured data and solving class imbalance. Combining these two algorithms, based on deep feature expression learning, and integrating multiple decision trees to solve the classification imbalance problem, has important practical significance for helping enterprises better understand customer needs and improve their competitiveness.

This article applies DNN and XGBoost algorithms, uses the transaction records of a clothing flagship store on an e-commerce platform as data objects, and performs a thorough analysis of consumer purchase decisions in financial and commercial big data in order to increase the accuracy of consumer behavior analysis and the scientificity of marketing strategies. From the results of accuracy and precision, compared with Logistic Regression (LR), Random Forest (RF), Support Vector Machine (SVM), Decision Tree (DT), and single DNN, the accuracy of this model is 15.8%, 10.5%, 13.0%, 14.4%, and 8.2% higher; the precision is 17.5%, 10.2%, 13.0%, 15.9%, and 10.3% higher; the AUC values are increased by 0.125, 0.076, 0.100, 0.114, and 0.044, respectively. In practical applications, combining DNN with XGBoost to analyze consumer behavior can help provide a basis for formulating marketing strategies.

2. Related Works

Consumer purchase decisions are the result of the combined action of multiple factors [4-5]. Current existing research focuses on solving the problems of data processing and pattern recognition. To accurately predict purchase behavior and provide personalized services to customers based on their preferences, Alghanam Orieb Abu proposed a data mining model to find association rules for frequent item sets. Dataset testing showed that the proposed model could effectively utilize consumer data and provide a basis for customer behavior prediction [6]. Lael Tri Ahmad Djabalul used the FP-Growth algorithm to conduct data mining prediction of consumer purchase patterns. The research results indicate that the FP Growth algorithm has the potential to detect consumer purchasing trends and provide useful data for enterprise marketing strategy planning and sales growth [7]. Existing research has enhanced the insight into market trends and improved the effectiveness of marketing strategies in a certain area. However, in the face of a rapidly changing market environment, its prediction accuracy is limited.

Due to its advanced development, artificial intelligence technology is increasingly being utilized to forecast consumer purchases [8-9]. Taking the ML model as an example, it has improved the ability to recognize diverse consumer behavior patterns and can provide support for improving prediction accuracy [10-11]. Wang Wenle suggested an XGBoost-based model for predicting user purchasing behavior. He analyzed multidimensional characteristics using the XGBoost feature significance model and determined that the model with the greatest weight value was the most important feature to employ when creating the model. According to the results, the suggested model demonstrated exceptional stability and algorithm efficiency, with an accuracy of 0.9761 when compared to the current ML model [12]. Li Yue presented a novel classifier for predicting consumer consumption behavior that is based on enhanced genetic extreme gradient boosting. It was evaluated against a number of popular parameter optimization algorithms and cutting-edge classification techniques. The outcomes demonstrated that the suggested approach outperformed the current approaches, and the prediction findings may serve as a basis for businesses to make better marketing decisions [13]. Existing research can effectively achieve accurate prediction of purchase behavior and assist precision marketing and recommendation, but there are still challenges in capturing the nonlinear relationship between consumer behavior factors and dealing with classification imbalance problems.

3. Methods

3.1 Data Collection and Preprocessing

This article takes users' shopping records, browsing records, and user comments as data objects, and analyzes users' consumption decision behaviors to explore users' consumption preferences and demand trends. First, shopping records and browsing records are obtained from an e-commerce platform. This article uses the standardized application programming interface provided by the platform to collect data including product sales volume, price and user information, and counts user click streams to analyze user purchasing behavior and patterns.

In terms of data processing, the collected data is cleaned to remove missing values and outliers. For missing data, the K nearest neighbor algorithm is used to infer missing data from the attribute values of similar samples:

$$\hat{x}_i = \frac{1}{k} \sum_{x_j \in N_k(x_i)} x_j \quad (1)$$

Here, $N_k(x_i)$ represents the k nearest neighbors closest to x_i .

For outliers, the box plot method is used to identify and eliminate extreme values. Based on the interquartile range, outliers are defined as lower than:

$$Q_1 - 1.5 \times IQR \quad (2)$$

or higher than:

$$Q_3 + 1.5 \times IQR \quad (3)$$

Among them, Q_1 and Q_3 are the first and third quartiles, respectively.

3.2 Feature Selection

The user's features are screened through the collected shopping records, browsing records, and user comment data. The Lasso regression method is used to evaluate the importance of each factor. A basic feature set of users and products is constructed, and the importance score $S(f)$ is calculated for each feature f . It is obtained by observing the absolute value of the regression coefficient:

$$S(f) = |\beta_f| \quad (4)$$

Among them, β_f is the regression coefficient corresponding to feature f .

One-hot encoding is used to transform categorical variables into a format that can be entered into a model. One-hot encoding, which is used to represent whether a categorical variable belongs to a particular category, is used to transform it into a binary variable under the assumption that it contains categories.

For numerical variables, normalization is used to ensure that each factor is on the same scale, and the Z-score normalization method is used. The calculation formula is:

$$z = \frac{x - \mu}{\sigma} \quad (5)$$

Among them, x is the original value; μ is the sample mean; σ is the standard deviation.

According to the feature importance score, the features with high scores are selected for analysis. The basic feature set of this article is shown in Table 1:

Table 1. Basic feature set

Sequence	Feature	Score
1	User ID	0.23
2	User gender	0.16

3	User age	0.09
4	Browsing frequency	0.12
5	Last purchase time	0.17
6	Product evaluation score	0.08
7	Product sales volume	0.14
8	Historical purchase amount	0.11
9	Repurchasing	0.10
10	Consumption level	0.15
11	Commodity price	0.21
12	Browse preferences	0.19

3.3 DNN Model Construction

Faced with the huge amount of complex financial and commercial big data, traditional analysis methods find it difficult to mine deeper information and nonlinear connections between these data. The multi-level structure and nonlinear activation function of DNN can effectively mine the deep association of financial data and improve the accuracy of consumer decision prediction.

DNN accepts the pre-processed feature set through the input layer, and then linearly combines the weight matrix and bias terms through multiple hidden layers. On this basis, nonlinear activation functions are used to apply complex patterns for learning, so as to better capture high-level interactions and complex nonlinear relationships [14-15]. This article uses DNN to analyze the nonlinear relationship between data features, as shown in Figure 1.

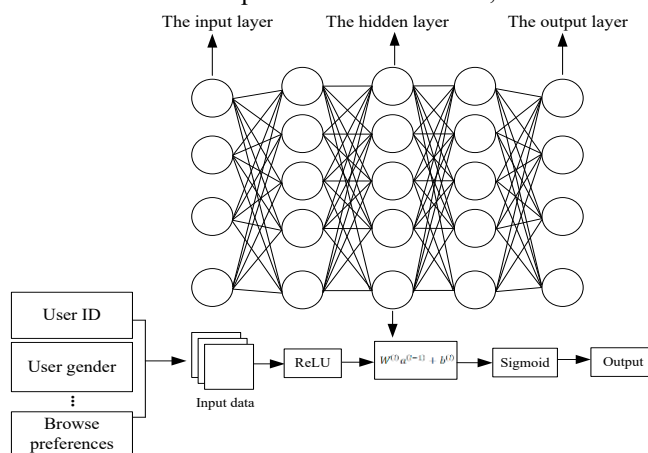


Fig 1. DNN model construction

First, the synthetic feature set for feature screening is used as the input training sample; assuming that the number of features is n , the input layer consists of n nodes. On this basis, a DNN model with multiple hidden layers is designed. The network is divided into L hidden layers, and each layer consists of m_l neurons ($l = 1, 2, \dots, L$). For each layer l , the input $a^{(l-1)}$ (effective output of the upper layer) is multiplied by the weight matrix $W^{(l)}$, and the bias term $b^{(l)}$ is added:

$$z^{(l)} = W^{(l)}a^{(l-1)} + b^{(l)} \quad (6)$$

Here, $z^{(l)}$ represents the unactivated output vector.

To apply nonlinearity, the ReLU (Rectified Linear Unit) activation function is applied:

$$a^{(l)} = \sigma(z^{(l)}) = \max(0, z^{(l)}) \quad (7)$$

On this basis, the output layer is established to produce the final prediction result. In the binary classification task, there is generally only one neuron in the output layer, and the Sigmoid activation function is used:

$$\hat{y} = \sigma(z^{(L+1)}) = \frac{1}{1 + e^{-z^{(L+1)}}} \quad (8)$$

Here, $\hat{y}$ is the probability estimate of the model when the sample is in the positive category.

3.4 XGBoost Integration

DNN has outstanding advantages in characterizing complex nonlinear relationships and high-order interactions, but it is not effective in solving classification imbalance problems. XGBoost integrates multiple decision trees together and has obvious advantages in processing imbalanced data. At the same time, the algorithm can also flexibly adjust the loss function to optimize the model performance. This article uses DNN to extract deep feature relationships from the comprehensive feature set, and feeds these features as input into the XGBoost model to further optimize the prediction accuracy.

The core of the XGBoost algorithm is its classification rules. It is an algorithm based on gradient boosting. Different from traditional algorithms, XGBoost is an incremental model, that is, decision trees are added step by step [16]. Each tree is based on the previous tree, and the remaining parts are trained, thus approaching the optimal solution step by step, as shown in Figure 2.

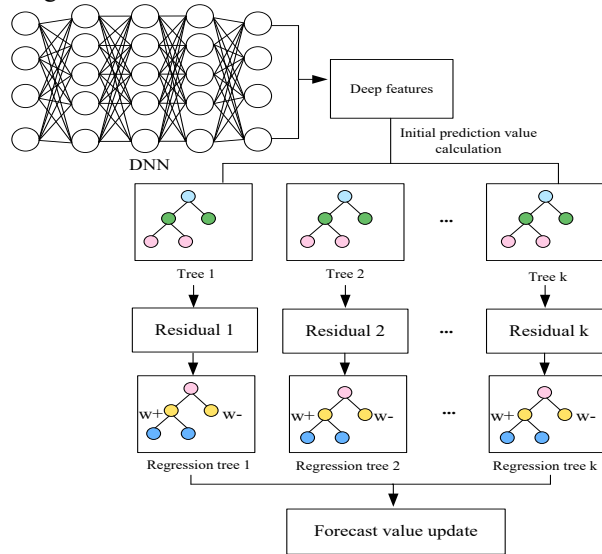


Fig 2. XGBoost model

The objective function of XGBoost consists of a loss function and a regularization term:

$$\text{Obj} = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k) \quad (9)$$

Here, $l(\hat{y}_i, y_i)$ is the direct loss function of the predicted value $\hat{y}_i$ and the true value y_i , and $\Omega(f_k)$ is the regularization term of the k -th tree, and is defined as:

$$\Omega(f_k) = \gamma T + \frac{1}{2} \lambda \|\omega\|^2 \quad (10)$$

Among them, T is the number of leaf nodes; γ and λ are hyperparameters, which are used to adjust the complexity and weight of the tree structure respectively. In this way, XGBoost can effectively solve the “overfitting” problem caused by the overly complex model while ensuring the prediction accuracy.

For consumption data with unbalanced classification, the sampling weighting method is used to achieve balance between categories. Assuming that in two different types of consumption decisions, the proportions of positive and negative samples are:

$$r = \frac{n_+}{n_-} \quad (11)$$

The weights of positive samples and negative samples are set to w_+ and w_- , respectively. In the specific implementation process of $w_+ \cdot n_+ = w_- \cdot n_-$, the sampling weighting is added to the loss function:

$$L_{\text{weighted}}(y, \hat{y}) = -w_+ y \log(\hat{y}) - w_- (1 - y) \log(1 - \hat{y}) \quad (12)$$

In ensemble learning, deep features are used as training samples to input the XGBoost model. Each sample point is preliminarily predicted, and a decision tree is established. First, the current prediction results and actual marks are used to calculate the residuals, and a new regression tree is constructed with the residuals as the control variable. On this basis, the sample points are divided into several leaf nodes, and each leaf node is weighted; on this basis, the prediction results of the new tree are combined with the existing predictions to obtain new prediction values. In this process, residual calculations and prediction value updates are continuously performed until the required number of trees reaches the required number or other termination conditions are met.

4. Results and Discussion

4.1 Experimental Settings

The dataset used in the experiment comes from the 30-day transaction records of a clothing flagship store on the e-commerce platform, including consumer purchase history, browsing history, and user evaluation information. According to the consumption behavior in the first 20 days, whether the product is purchased in the next 10 days is predicted. If there is a purchase behavior in the next 10 days, it is a positive sample, otherwise it is a negative sample. A total of 121,517 samples—17,632 positive samples and 103,885 negative samples—are obtained following data cleaning and feature screening preprocessing.

Three broad assessment criteria are chosen in order to thoroughly assess the model's performance:

- (1) Accuracy: the proportion of all samples that were correctly predicted.
- (2) Precision: the proportion of samples that are truly positive compared to those that were anticipated to be positive.
- (3) AUC: The model's capacity to differentiate between positive and negative samples is gauged by the area under the Receiver Operating Characteristic (ROC) curve.

Three standard assessment metrics are chosen in order to thoroughly assess the model's performance:

The model in this article is compared with five approaches in the analysis of experimental findings to confirm the efficacy of the approach in this article: LR, RF, SVM, DT, and single DNN model:

The algorithm parameter settings of this article are shown in Table 2:

Table 2. Algorithm parameter settings

Classification	Parameter	Setting
DNN	Number of hidden layers	3
	Number of neurons	128, 64, 32
	Learning rate	0.001
	Batch size	32
	Activation function	ReLU

XGBoost	Regularization term	L2, $\lambda = 0.001$
	The number of trees	100
	Maximum depth	6
	Learning rate	0.1

Other comparison models use standardized parameter settings.

4.2 Experimental Results

(1) Accuracy and Precision

Based on the dataset, this article compares the accuracy and precision results of each method, as shown in Figure 3:

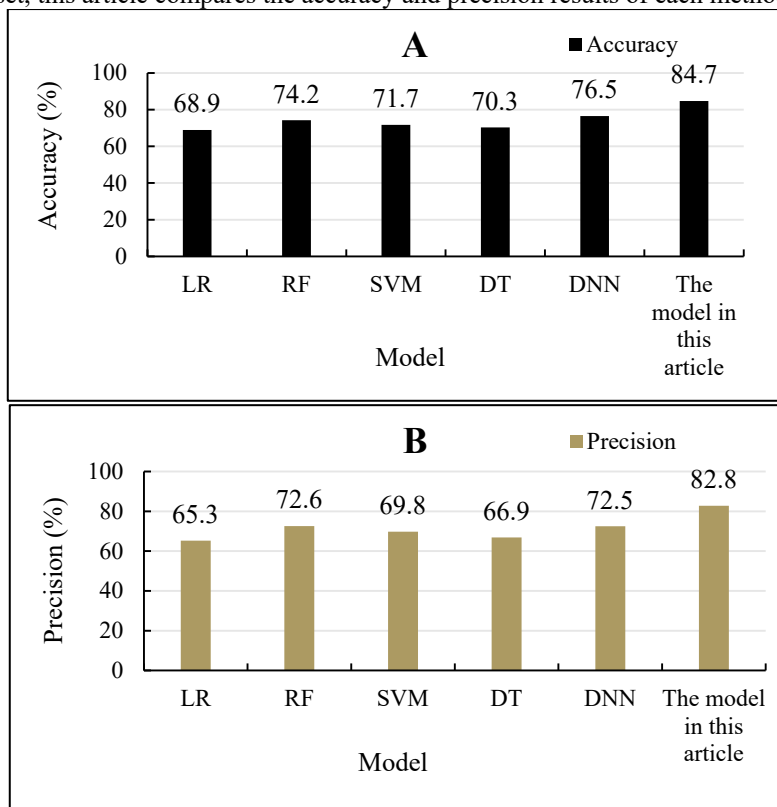


Fig 3. Accuracy and Precision comparison

Figure 3A is Accuracy

Figure 3B is Precision

From Figure 3, the model of this article has a more significant advantage in the comparison of Accuracy and Precision. In Figure 3A, the accuracy of the model of this article in predicting consumer purchase behavior in the dataset is 84.7%, and the accuracy of LR, RF, SVM, DT, and single DNN models are 68.9%, 74.2%, 71.7%, 70.3%, and 76.5%, respectively. In comparison, the accuracy of the model of this article is 15.8%, 10.5%, 13.0%, 14.4%, and 8.2%, higher. In Figure 3B, the precision of the proposed model reaches 82.8%, while the precision of LR, RF, SVM, DT, and single DNN models are 65.3%, 72.6%, 69.8%, 66.9%, and 72.5%, respectively. Compared with them, the precision of the proposed model is 17.5%, 10.2%, 13.0%, 15.9%, and 10.3% higher.

The proposed model uses DNN to extract deep feature associations, which enables it to have a deeper understanding of the inherent laws of consumer behavior. On this basis, the XGBoost algorithm is used to achieve

accurate prediction of multi-class data, reduce the false alarm rate, accurately predict multi-class data, and achieve efficient recognition of small samples.

(2) AUC

AUC offers a thorough assessment of the model's capacity to differentiate between positive and negative data. It can more precisely represent the model's performance at various thresholds and is unaffected by class imbalance. The AUC results of each model method are shown in Figure 4:

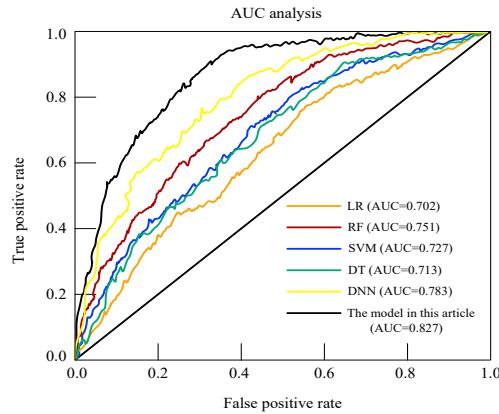


Fig 4. AUC comparison

From Figure 4, the AUC results of this article are more ideal. In the specific results, the AUC of this model reaches 0.827, and the AUC values of LR, RF, SVM, DT, and single DNN models are 0.702, 0.751, 0.727, 0.713, and 0.783, respectively. The AUC values of this model are improved by 0.125, 0.076, 0.100, 0.114, and 0.044, respectively, when compared to the comparison model.

This outcome demonstrates how well the model in this article can distinguish between positive and negative samples. By utilizing the multi-level nonlinear transformation of DNN, it can effectively mine the complex laws and advanced interactions of consumer behavior, so as to better identify potential consumer signals. Deep feature extraction can effectively improve the discriminability of the model for multi-class samples and improve AUC; the XGBoost algorithm increases the weight of small class samples to ensure the importance and rarity of small samples, thereby improving the overall discriminability of the model. LR assumes that the characteristics of consumer behavior are linearly related to the measured variables, and cannot effectively characterize the nonlinearity and high-order interactions of consumer behavior, making it difficult to effectively discriminate them. The DT model has the disadvantage of overfitting when dealing with nonlinear relationships. Although RF and SVM have great advantages in generalization and classification, they are prone to redundant operations or underfitting for high-dimensional sparse data. Although a single DNN can extract deep features, it lacks an effective method to deal with the problem of imbalanced classification, and is prone to bias against samples of most categories, which in turn affects its ability to identify small samples. In comparison, the model combining DNN and XGBoost in this article has obvious advantages.

4.3 Discussion

This article uses DNN and XGBoost algorithms to study the precision improvement of consumer purchase decisions in big data. Through experimental analysis, the effectiveness of this model is verified. By combining DNN and XGBoost technology, the complexity and imbalance of consumer decision behavior can be effectively solved. DNN can effectively mine complex patterns and advanced interactions in data, while XGBoost has the advantages of integrated learning, especially class imbalance problems, thereby improving the discriminability and generalization ability of the model. This article provides a reference for solving the problem of data classification imbalance, and also provides certain guidance for the precision marketing of enterprises.

5. Conclusions

In view of the complex prediction of consumer behavior and class imbalance under financial and commercial big data, this article uses DNN and XGBoost algorithms to analyze consumer purchase behavior in financial and commercial big data. Based on DNN, deeper features are mined from big data. The loss function and sample weights are optimized by XGBoost method to improve the ability to analyze consumer decisions. The experimental results show that the accuracy of this model reaches 84.7%; the precision reaches 82.8%; the AUC value reaches 0.827, which is significantly better than the control model. The combination of DNN and XGBoost models can not only better characterize complex nonlinear relationships, but also better improve the accuracy of predictions in the case of unbalanced classification. This article can provide a reference for consumers' consumption decisions and precision marketing. However, there are still some shortcomings. When training large data based on the combination of DNN and XGBoost, the amount of calculation required is very large, and the process of adjusting parameters is relatively complicated. Future research can consider further optimizing the structure and parameters of the model and reducing computing consumption under the premise of ensuring high prediction effects, so as to provide a basis for high-quality marketing of enterprises in a digital environment.

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